

The Two Prices of Dependence: a Vanishing Coverage Tax and a Fixed Sharpness Rent

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July 2026

Abstract

Temporal dependence charges split conformal prediction two different prices, and they scale differently. The coverage price is a tax that vanishes with the calibration size: for a stationary β -mixing score process the loss is at most $\min_{\tau}\{\tau/(n+1) + 2\beta(\tau)\}$ (Barber and Pananjady, 2026). The sharpness price is a rent that does not: the per-step log-score regret of the marginally calibrated predictive against the past-conditional oracle equals the entropy-rate gap $\Delta H = H(S_0) - h$, the mutual information between the present score and its infinite past, a constant independent of n . For Gaussian score processes the rent exponentiates into width — oracle intervals are narrower by the factor $e^{-\Delta H}$ at matched coverage, which for an AR(1) score process with parameter ϕ is $\sqrt{1 - \phi^2}$. Dependence is therefore asymptotically free for the certificate and permanently valuable for the forecast.

1 Setting

Let $(S_t)_{t \in \mathbb{Z}}$ be a stationary ergodic sequence of conformity scores with continuous marginal distribution $\bar{F}$ and marginal density $\bar{f}$. Write $\mathcal{P}_t = \sigma(S_{t-1}, S_{t-2}, \dots)$ for the past, $F_t = \mathcal{L}(S_t | \mathcal{P}_t)$ for the past-conditional law with density $f(\cdot | \mathcal{P}_t)$, $H(S_0)$ for the differential entropy of the marginal, and $h = \lim_{m \rightarrow \infty} H(S_0 | S_{-1}, \dots, S_{-m}) = H(S_0 | \mathcal{P}_0)$ for the entropy rate (Cover and Thomas, 2006). All entropies are assumed finite. Two predictive systems are compared at each step:

- the *marginal* system, which issues the stationary law $\bar{F}$ (this is what split conformal calibrates: its quantile is an estimate of a quantile of $\bar{F}$, and the predictive density it implies is the one residual shape $\bar{f}$, re-levelled);
- the *past-conditional oracle*, which issues F_t .

The comparison isolates what marginality itself forfeits; estimation error in either system is not the subject.

2 The coverage tax

Dependence costs the marginal system coverage, and the cost is quantified by Barber and Pananjady (2026): for a stationary β -mixing score sequence, split conformal with n calibration points and a memoryless pretrained score satisfies

$$\mathbb{P}(\text{cover}) \geq 1 - \alpha - \min_{\tau \geq 0} \left\{ \frac{\tau}{n+1} + 2\beta(\tau) \right\}, \quad (1)$$

with a matching upper bound, and the bound is tight up to a constant factor over this class. For fixed dependence the tax vanishes: for a process whose dependence dies at lag t (e.g. MA(t)), the loss is at most $(t+1)/(n+1)$, and for geometrically β -mixing processes (e.g. stationary Gaussian AR(1)) it is $O(\log n/n)$. The earlier rate of Oliveira et al. (2024), of order $\sqrt{\tau/n}$, is likewise vanishing. Nothing in this section is new; it is quoted to fix one side of the comparison.

3 The sharpness rent

Proposition 1 (Rent identity). *The per-step expected log-score regret of the marginal system against the past-conditional oracle is*

$$\mathbb{E}\left[\log \frac{f(S_0 | \mathcal{P}_0)}{\bar{f}(S_0)}\right] = \mathbb{E} \text{KL}(F_0 \| \bar{F}) = I(S_0; \mathcal{P}_0) = H(S_0) - h =: \Delta H.$$

Proof. The first equality is the tower property. The second is the standard identity $I(X; Y) = \mathbb{E}_Y \text{KL}(\mathcal{L}(X | Y) \| \mathcal{L}(X))$ applied with $X = S_0, Y = \mathcal{P}_0$. The third is $I(S_0; \mathcal{P}_0) = H(S_0) - H(S_0 | \mathcal{P}_0)$ and the definition of the entropy rate of a stationary process. $\square$

The rent is a property of the score process alone. It does not depend on the calibration size n , and no re-levelling of the marginal shape can reduce it: conformalizing acts on $\bar{f}$ by a monotone re-levelling, which leaves every likelihood ratio $f(\cdot | \mathcal{P}_0)/\bar{f}(\cdot)$'s expectation unchanged (Cotton). Proposition 1 is the identity of that paper with the side information taken to be the past rather than a covariate; the transfer requires stationarity and finite entropies, nothing else.

Corollary 1 (Gaussian processes: rent as width). *Let (S_t) be stationary Gaussian with marginal variance σ^2 and one-step innovation variance σ_∞^2 (the Kolmogorov–Szegő prediction variance). Then*

$$\Delta H = \frac{1}{2} \log \frac{\sigma^2}{\sigma_\infty^2},$$

and at any matched coverage level the past-conditional oracle's central interval is narrower than the marginal interval by the factor $\sigma_\infty/\sigma = e^{-\Delta H}$. For an AR(1) process with parameter ϕ this factor is $\sqrt{1-\phi^2}$ and $\Delta H = -\frac{1}{2} \log(1-\phi^2)$.

Proof. For Gaussian laws $H = \frac{1}{2} \log(2\pi e \sigma^2)$ and $h = \frac{1}{2} \log(2\pi e \sigma_\infty^2)$; both central intervals are z -multiples of the relevant standard deviation at matched level. For AR(1), $\sigma^2 = \sigma_\infty^2/(1-\phi^2)$. $\square$

4 The asymmetry

Fix a stationary Gaussian AR(1) score process and let n grow. The tax (1) is $O(\log n/n)$; the rent is $-\frac{1}{2} \log(1-\phi^2)$ at every n . Numerically:

ϕ	rent ΔH (nats)	oracle width / marginal width
0.3	0.047	0.954
0.6	0.223	0.800
0.9	0.830	0.436

At $\phi = 0.9$ the marginal certificate costs essentially nothing to keep — a tax of order $\log n/n$ — while the forecaster who conditions on the past issues intervals less than half as wide at the same coverage, forever. The two prices answer different questions. The tax answers: how much coverage

does the certificate lose because the calibration data are dependent? The rent answers: how much sharpness does the forecast lose because the calibration ignores that dependence? The literature on conformal prediction for time series has concentrated on driving the first price to zero; the second price is the one a forecaster pays at every step, and it is not reduced by any amount of calibration data.

5 Limits

Three restrictions delimit the claim. The comparison is between the marginal system and the oracle, so it prices marginality itself; a feasible past-conditional model pays estimation costs the oracle does not, and under strong dependence those costs can be material at small n . The width statement of Corollary 1 is Gaussian; for general processes the rent is the log-score statement of Proposition 1, and width comparisons depend on the shape of the conditional laws. And the rent as stated prices the score process; when scores are residuals from a fixed predictor, conditioning information in the original series that the predictor already absorbs does not appear in ΔH . Within these limits the accounting is exact, and the direction of the asymmetry does not depend on the Gaussian case: the tax vanishes in n and the rent does not.

References

- R. F. Barber and A. Pananjady. Predictive inference for time series: why is split conformal effective despite temporal dependence?, 2026. Preprint, arXiv:2510.02471.
- P. Cotton. Marginally useful: Formalizing the information gap in conformal prediction. Companion paper.
- T. M. Cover and J. A. Thomas. *Elements of Information Theory*. Wiley, 2nd edition, 2006. Chapter 6, gambling and side information; Pinsker’s inequality.
- R. I. Oliveira, P. Orenstein, T. Ramos, and J. V. Romano. Split conformal prediction and non-exchangeable data. *Journal of Machine Learning Research*, 25, 2024.